

Kimi K3 vs Grok 4.5, Claude Fable 5 & GPT-5.6 Sol

Moonshot's 2.8T open-weight flagship, at its single shipped effort level, against the Report 07 field: twenty-three real merged post-cutoff PRs across Python, Rust, TypeScript, JavaScript, and Go, graded by deterministic hidden tests.

Abstract

Kimi K3 ships one effort level (max, always-thinking): 17/23 (74%) at the standard 30-minute budget, 20/23 (87%) at two hours — level with Fable 5 and Sol at high effort, at \$1.37 vs. \$0.80–\$1.04 per solved task. Five of its six capped misses were timeouts (the sixth scored 0.7 partial credit); at 18.1 min/task it runs 4–7× slower than the field. Kimi benchmark spend \$33.89 under a \$40 cap.

Table 1. Results (Grok/Fable/Sol reproduced from Report 07)

Model (effort)	Score	pass@1	Cost	Tok/task	Time/task	\$/solved
Grok 4.5 (medium)	21/23	91.3%	\$6.67	106 K	3.9 min	\$0.32
Grok 4.5 (high)	21/23	91.3%	\$8.76	141 K	4.9 min	\$0.42
Claude Fable 5 (low)*	20/23	87.0%	\$12.18	28 K	3.4 min	\$0.61
GPT-5.6 Sol (high)	20/23	87.0%	\$15.90	85 K	4.2 min	\$0.80
Claude Fable 5 (high)*	20/23	87.0%	\$20.82	44 K	4.3 min	\$1.04
Kimi K3 (max, 2-h budget)‡	20/23	87.0%	\$27.43	216 K	28.3 min	\$1.37
Grok 4.5 (low)	19/23	82.6%	\$3.39	57 K	2.5 min	\$0.18
GPT-5.6 Sol (medium)	19/23	82.6%	\$8.83	50 K	2.8 min	\$0.46
Claude Fable 5 (medium)*	19/23	82.6%	\$16.32	36 K	3.7 min	\$0.86
GPT-5.6 Sol (low)	18/23	78.3%	\$3.85	23 K	1.5 min	\$0.21
Kimi K3 (max, 30-min budget)	17/23	73.9%	\$16.84	134 K	18.1 min	\$0.99

‡ 2-hour row: the five tasks that timed out at 30 minutes were re-run with exact budgets of 2 h / 400 steps / \$4 per run (runs marked `budget_override`); the other 18 runs unchanged. It is an ablation, not a leaderboard config — every other model ran under the standard budget. * Fable 5 ran with Opus 4.8 refusal fallback (see Report 07). Tokens are billed tokens; time is sandbox wall-clock.

Findings

1. The clock is the dominant failure mode — not the only one.

Five of the six capped misses were timeouts mid-work; the sixth (SQLGlot ISO-8601) was a submitted patch that earned 0.7 partial credit. Given two hours, three timeouts convert, while the two hardest tasks still end at the \$4 cost cap with partial progress.

2. Always-on max thinking is slow, not verbose.

18.1 min/task against 3.4–4.9 for the field, on moderate token volume (134 K/task). The cost of Preserved Thinking is time; on time-budgeted agentic work, latency is accuracy.

3. Given time, it closes the gap.

At 2 hours, aiohttp (64 min), NetworkX-Leiden (107 min), and jiff-strftime (21 min — run-to-run variance, not time starvation) all fall: 20/23, matching high-effort Fable 5 and Sol for +\$10.59.

4. The difficulty ceiling held.

PennyLane Trotter fragmentation (0.2 partial) and SQLGlot internal-name canonicalization (0.17 partial) remain unsolved by every configuration ever run on v3 — 0-for-27 in Report 07, and K3's two-hour, \$4 attempts didn't crack them either.

Budget ablation (what the extra ninety minutes bought)

Task	30-min budget	2-h budget	Time	Cost
aiohttp-upgrade-deferred	x timeout	ok solved	64 min	\$2.66
networkx-leiden-communities	x timeout	ok solved	107 min	\$4.35
jiff-strftime-negpad	x timeout	ok solved	21 min	\$1.14
pennylane-trotter-fragmented	x timeout	partial (0.2), \$4 cap	100 min	\$4.78
sqlglot-canonicalize-internal-names	x timeout	partial (0.17), \$4 cap	94 min	\$4.12
sqlglot-iso8601-nanos	partial (0.7)	— not re-run	—	—

The two \$4-capped partials ended on money, not conviction — a larger budget might buy more progress, at a meaningfully higher price than any other configuration on the board.

Method & reproducibility

Same 23 v3 tasks as Report 07 (real merged post-cutoff PRs: Python 9, Rust 4, TypeScript 4, JavaScript 3, Go 3). Network-isolated Docker sandboxes; deterministic hidden-test grading; one attempt per task per configuration. Every task pre-validated: gold patch = 1.0, unpatched = 0.0, deterministic over 3 runs. kimi-k3 via the Moonshot API (\$3/\$15 per million tokens, cache-hit input billed here at the cache-miss rate); thinking is always on and reasoning_effort accepts only "max" (further levels announced but unshipped), so K3 contributes one effort configuration. Harness note: three solves were initially lost to 120-second judge-call read timeouts, recovered with a 900-second provider timeout. Reproduce: `vulcanbench run --suite v3 --model kimi:kimi-k3 --effort extra-high` — the ablation adds `--tasks-root tasks/v3 --timeout 7200 --max-steps 400 --override-budgets --max-run-cost 4` per task.